

Gwyddion: an open-source software for SPM data analysis

Research Article

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Abstract:

In this article, we review special features of Gwyddion—a modular, multiplatform, open-source software for scanning probe microscopy data processing, which is available at <http://gwyddion.net/>. We describe its architecture with emphasis on modularity and easy integration of the provided algorithms into other software. Special functionalities, such as data processing from non-rectangular areas, grain and particle analysis, and metrology support are discussed as well. It is shown that on the basis of open-source software development, a fully functional software package can be created that covers the needs of a large part of the scanning probe microscopy user community.

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1. Introduction

Scanning probe microscopy (SPM) is a well established experimental method for determining the surface morphology and other local physical quantities with very high resolution. The range of SPM experiments found in the

present-day literature extends from atomically resolved imaging of surface reconstructions [1, 2] to long-range measurements of critical dimensions in the semiconductor industry [3, 4]. Many particular variants of SPM are known, including possibilities of local morphological measurements and local force mapping (AFM – atomic force microscopy), magnetic measurements (MFM – magnetic force microscopy), thermal measurements (SThM – scanning thermal microscopy), optical measurements (SNOM – scanning near-field optical microscopy), as well as a number of other techniques.

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Substantial efforts and costs are expended by researchers and instrument manufacturers for the development of novel and better SPM instruments. Many commercial instruments can reach noise levels far in the sub-atomic range in all three motion axes. The influence of data processing in SPM measurements is, however, often underestimated, and data processing is not considered when discussing SPM measurement accuracy.

SPM data represent only an approximation of the local surface morphology or local physical properties. Even during the measurement, SPM data are regularly sampled, which can be understood as a spatial filtering technique. Moreover, SPM only rarely provides information that can be evaluated without any further data processing. Usually, at least basic leveling routines need to be applied to remove the tilt between the axes of the microscope and the mean sample plane. SPM data can also be heavily influenced by the probing model, e.g., the tip-sample convolution in AFM measurements [5]. These effects need to be taken into consideration at the data-processing stage. Finally, high sensitivity does not necessarily mean high precision, and proper calibration routines need to be applied as well [6].

Unfortunately, SPM data processing is a much weaker part of SPM data-acquisition process than the measurement itself. Algorithms used for data processing are typically part of the built-in software provided by the instrument manufacturer and are therefore often unknown. Many algorithms are based on user input only, relying solely on user experience. Data-processing repeatability can therefore be quite poor. Thus, it is important to employ reliable data-processing software and to select algorithms minimizing user influence.

There are several projects aimed at developing a universal, SPM manufacturer independent software for SPM data analysis on a commercial¹, a non-commercial [7], or an open-source basis [8]. In this article, we review the open-source SPM data-processing software Gwyddion, which is available free of charge at². We show that it provides a number of special tools for quantitative data processing in SPM. Since Gwyddion is available for various platforms, it can reach a large part of the SPM user community. Moreover, Gwyddion software libraries can easily be integrated into any other program, such as data-acquisition software. This can substantially simplify both data acquisition and processing in many other situations, e.g., by writing custom software for SPM instrument control.

2. Gwyddion description

Gwyddion is a modular multiplatform software for SPM data analysis. Its main task is to provide fast and reproducible data-processing routines for SPM users. It allows users to perform all basic operations necessary for a successful evaluation of the data obtained using an SPM. The extensive list of features (see³) contains, for example, these basic and most commonly used ones:

- Basic visualization of SPM data: false-color representation, including shading, edge detection, and similar visualization tools.
- OpenGL 3D data display: false-color or material representation, easily editable color maps, and OpenGL materials.
- Multiple-channel treatment: data arithmetic and selection of multiple channels.
- Instrument tilt or non-planarity compensation: plane leveling, profiles leveling, three-point leveling, facet leveling, polynomial background removal, leveling along user-defined lines.
- Extraction and treatment of profiles: profile for arbitrary thickness extraction, distances in profile graph, profile export and simple graph-function fitting, critical-dimension determination.
- Data filtering: mean, median, conservative denoise, Kuwahara, minimum, maximum, checker pattern removal, general convolution filter with user-defined kernel.
- Basic statistical functions and quantities: roughness parameters, 1D and 2D correlation functions, power spectral-density function, angular distributions.
- ISO roughness parameter evaluation.
- Special statistical functions: fractal-dimension analysis, 2D slope and facet analysis, Minkowski functionals.
- Integral transforms: 1D and 2D FFT filtering, wavelet filtering.
- Correction of image defects: spot removal, outlier marking, scar (stroke) marking, line correction

¹ <http://www.imagemet.com/>

² <http://gwyddion.net/>

³ <http://gwyddion.net/features.php>

methods, removal of data under an arbitrary mask using Laplace or fractal interpolation, fast scan axis drift correction.

- Support for tiled imaging: merging and immersion of images.
- Basic probe-sample interaction modeling using a geometrical approach: tip modeling, blind estimation, dilation, and erosion.

Besides the basic functionality listed above, there are several special features that should help spreading Gwyddion among SPM users:

- Gwyddion is available for all common operating systems.
- Gwyddion can read over 80 file formats, which represents the absolute majority of all files formats produced by SPMs in the world.
- Most of the basic functionality is in libraries with stable API and modules that can be used for the construction of other SPM data-acquisition or processing software.
- All features, including API for all library functions, are documented on-line.
- Being open source and having a clear library and data-processing modules structure, anyone can inspect or improve the Gwyddion code.

In this article, we primarily describe the main attributes that set Gwyddion apart from other SPM data-processing software.

2.1. Modularity

Gwyddion itself is an extremely small application. The binary representation of the Gwyddion executable occupies less than 2 percent of the full installation, while the rest are Gwyddion libraries and data-treatment modules. Gwyddion is written in C and builds on Gtk+ and GLib libraries. It relies on the GLib utility library for portability and uses the GLib object system GObject for its own objects. The graphical user interface is implemented with the Gtk+ toolkit with a fair amount of Gwyddion-specific extension widgets. The program can be divided into the following main components:

- libraries providing basic and advanced data-processing routines, graphical user interface elements, and other utility functions and objects,

- the quite small and simple actual application serving primarily as a glue connecting the other components in a common graphical interface, and
- modules (technically: run-time loaded libraries) that provide most of the actual functionality and present it to the user; in many situations, they extensively use library methods.

Libraries are the key part of the provided software, and these are also components that can easily be integrated into any other software. The basic Gwyddion library structure is as follows:

- The `libgwyddion` library defines some core interfaces, like serialization (`GwySerializable`) for data-like objects; general classes not directly related to data processing or GUI (hash-table-like data container `GwyContainer`, general unit representation `GwySIUnit`); and various string, mathematical, file, and miscellaneous utilities and macros.
- The `libprocess` library defines two basic objects: `GwyDataField` representing two-dimensional data, and `GwyDataLine` representing one-dimensional data. These objects encapsulate raw floating-point arrays together with basic metadata—physical dimensions and their units. It provides fundamental one- and two-dimensional data-processing functions used subsequently as basic building blocks for specific functions in modules.
- The `libdraw` library is a small library that provides a few color-handling and elementary data-rendering functions. It defines false-color map objects (`GwyGradients`), OpenGL material objects (`GwyGLMaterials`), and the abstract selection class `GwySelection`.
- The `libgwydgets` library is essentially a collection of Gwyddion-specific Gtk+ widgets. There are several notable groups of widgets, namely 1D, 2D, and pseudo 3D data-representation views.
- The `libgwymodule` library deals with module loading and other administrative tasks, and it acts as a proxy for the usage of module functions. It defines functions for the registration of modules and their functions, and it also implements some more convenient wrappers around primitive module functions.
- The `libgwyapp` library contains functions related to the main application, namely application-level data management. Some of the functions could have

probably been equally well put into the main application. Nevertheless, many can be useful also outside Gwyddion itself—in a Gwyddion related program. As it is not always clear which are which, the library contains everything that may be of any interest to the outside world.

While the libraries contain the basic algorithms and data structures, Gwyddion modules provide most of the software interaction with the user (even if they can be run also in non-interactive mode). Modules can provide different types of functionality; from the viewpoint of user interaction, the following types are important:

- Data processing methods—functions that carry out some operation on given data.
- File loading and saving—data file type detection, loading, and saving functions that get a file name to load or detect, or a file name and data to save.
- Interactive tools—objects representing interactive tools that are selected from a tool palette and used in data windows, similar to how tools in some drawing programs operate.
- Graph methods—functions operating (allowing the user to operate interactively) on a given graph.

Everything besides the executable binary can be employed for any other purposes than running Gwyddion, e.g., for the development of other SPM data-acquisition or analysis software that fits user needs. Since the Gwyddion libraries have been developed with their possible use in other software in mind and support pkg-config⁴ (which is the standard mechanism for obtaining information on how to compile and link programs with an external library), their use in custom software is straightforward.

There is a large variety of software that can be created using Gwyddion libraries. Examples (some of them available to the public free of charge) are:

- Gwyview—a simple SPM data viewer, which also serves as an example of how to develop Gwyddion-based programs. Thanks to the use of Gwyddion loading and visualization functions, it can display any SPM file that can be opened in Gwyddion.
- GrapherAD—a data-visualization component of a custom software for fitting ellipsometric and spectrophotometric data developed by newAD and

used by the Department of Physical Electronics, Masaryk University. It employs primarily the Gwyddion graph widgets and also some infrastructure functions.

- GwyScan—a software using all Gwyddion libraries and some hardware (National Instruments cards, Atmel AtXmega microprocessors) for driving the metrology scanning probe microscope [9] and a long-range scanning probe microscope at the Czech Metrology Institute. Here Gwyddion is used for both data treatment and visualization, and the acquisition code for the entire microscope is limited to a few hundred lines.
- GSvit—software using the libprocess library for data handling in graphics card based on Finite Difference in Time Domain solver (see⁵). Here Gwyddion is used for platform-independent 1D and 2D data loading and saving.

2.2. Statistical data processing using masks

Statistical analysis is used frequently in SPM, for example when evaluating parameters of surface roughness. There are many statistical functions and quantities implemented in Gwyddion, which (from a quantitative point of view) do not differ from other SPM data-processing software. What does make a difference is the possibility of calculating statistical parameters from arbitrary non-rectangular selections that can be chosen by the user or by some segmentation algorithm.

Arbitrarily shaped subsets of pixels, marked by so called masks, are used as a universal mechanism for marking specific regions in the data, regardless of whether they represent grains, defects or regions of interest marked by the user. Subsequent operations can then take the mask into account and focus on the marked area. The separation of marking and processing provides great freedom and permits the user to combine these steps in new and unforeseen ways.

The mask creation and modification operations include:

- marking of grains, facets of a given slope, or defects, such as outliers or scars/strokes on the basis of the watershed algorithm [10];
- manual mask-creation editing with a mask editor that permits the user to draw rectangles, ellipses, or free-form shapes;

⁴ <http://pkgconfig.freedesktop.org/>

⁵ <http://gsvit.net/>

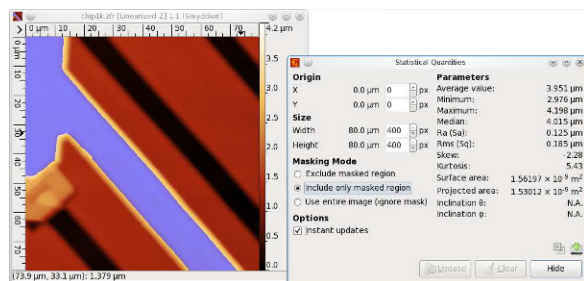


Figure 1. Gwyddion screenshot with the AFM image of a microchip surface and statistical quantities evaluated from masked rough aluminum parts.

- global operations, such as inversion, shrinking, or growing; and
- marking of data based on the values of some other data.

Masks can also be created as an auxiliary output of some operations to distinguish data with certain properties, e.g., surface reconstruction using a tip model marks areas with possible multiple tip touches or cross-correlation marks areas with low correlation score.

Operations with masked data can be divided into two main groups: grain or particle operations characterizing individual parts of the mask—these are discussed in Sec. 2.4—and application of the mask to include or exclude the corresponding data. The second group includes the calculation of statistical quantities (minimum, maximum, mean, rms, surface area, global curvature, ...), leveling (plane, facet, or polynomial), and interpolation and arithmetic operations with data. An example for the latter is the possibility to combine the pixels of two images 1 and 2 based on the mask on a third image 3 using the expression $d1*m3+d2*(1-m3)$ in the data arithmetic module.

As an illustration, the mask created on AFM data representing a microchip surface is shown in Fig. 1 together with the evaluated statistical roughness results. The mask, selecting aluminum interconnections in the microchip, was obtained using a combination of thresholding and manual selection. Similarly, additional data, e.g., representing the local thermal conductivity could be used for creating this selection.

2.3. Data synthesis

SPM measurements are often highly affected by the SPM probing mechanism referred to as tip-sample interaction. For morphological measurements using rigid bodies (both tip and sample), this effect is often described as tip-sample convolution, and we can simulate, evaluate, or correct tip-

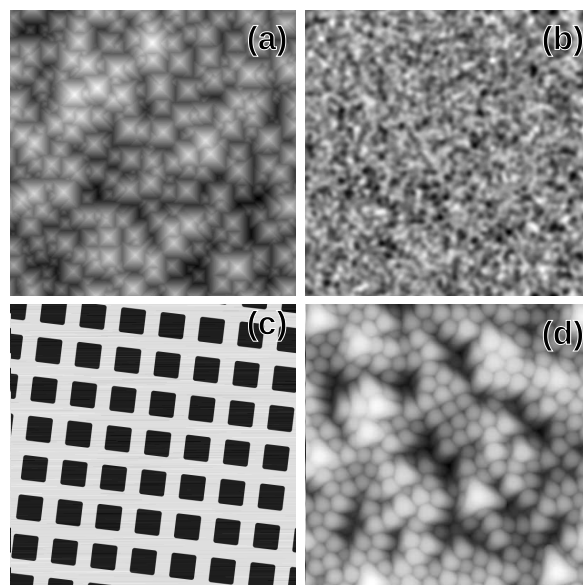


Figure 2. Examples of synthetic surfaces: (a) randomly patterned pyramidal surface similar to the top surfaces of crystalline silicon solar cells (b) randomly rough Gaussian surface (c) regular hole pattern with artificial line noise (d) simulated agglomerated nanoparticles on rough substrate, convolved with an AFM tip.

sample artifacts in such cases [5]. For the treatment of non-rigid bodies or measurements of other physical quantities, the situation is much worse—the tip-sample point-spread function is dependent on many variables and much deeper physical knowledge is needed to eliminate tip-sample artifacts. Simulating surfaces with given properties, performing some virtual measurements on it (e.g., by a dilation algorithm in the simplest case), and analyzing simulated data can yield a better understanding of real SPM images [15].

Gwyddion includes many algorithms for the simulation of both typical surfaces and instrumental defects, namely noise, that include:

- Random roughness with Gaussian or power-law autocorrelation functions can be generated using spectral synthesis.
- Nanoparticles on a surface can be simulated using a simple model similar to molecular-dynamics calculations. The aim of the model is to simulate the process of nanoparticle deposition and agglomeration often observed in real measurements.
- Common regular structures, such as steps, ridges, or hole patterns, can be generated, possibly with lateral deformations and variations of dimensional parameters, such as height or distance.

- Randomly patterned surfaces formed by randomly distributed surface features of a specific shape, e.g., pyramid, half-sphere, tetrahedron, or Gaussian, can be modeled using an object-placement algorithm.
- Finally, uncorrelated point noise and line noise can be generated with several one-sided and two-sided distributions, namely Gaussian, exponential, triangular, and uniform, to simulate noisy data.

All methods can not only be used to generate a purely artificial surface but also for modifications of existing surfaces. An illustration is presented in Fig. 2 showing just a few of the large number of possible combinations of different synthesis module results.

2.4. Particle and grain analysis

Particle analysis is an important challenge in present-day nanoscale metrology. Micro- and nanoparticles are used in many fields of research and technology. They can be characterized using different methods including optical [11] and x-ray scattering [12] or using different microscopies [13]. Nanoparticles are also often measured using AFM [14]. For isolated nanoparticles on flat substrates, this is a relatively easy task. However, in practice it is often necessary to analyze nanoparticles on rough substrates or nanoparticles that are not isolated. Using Gwyddion's data synthesis modules, we can simulate these effects, as shown in the previous section.

For particle and grain analyses, there are a number of algorithms available in Gwyddion. Both particles and grains are handled using a mask that marks areas that belong to individual grains or particles. Data processing therefore involves usually two steps:

1. Particles or grains are marked, i.e., a mask is created using some of the aforementioned algorithms.
2. Grain/particle statistical functions are used for mask analysis including the possibility of measuring many different quantities associated with the particle's projected size, boundary, volume, or orientation. High aspect-ratio nanoparticles can be analyzed as well. Results can be evaluated for both individual particles selected by the user or the entire ensemble of particles marked by a mask.

As an example, the statistical processing of agglomerated polymer nanoparticles on silicon is illustrated in Fig. 3. In this case, the watershed algorithm was used to mark the nanoparticles, and the resulting mask was evaluated using a particle equivalent projection algorithm to obtain the distribution of nanoparticle radii.

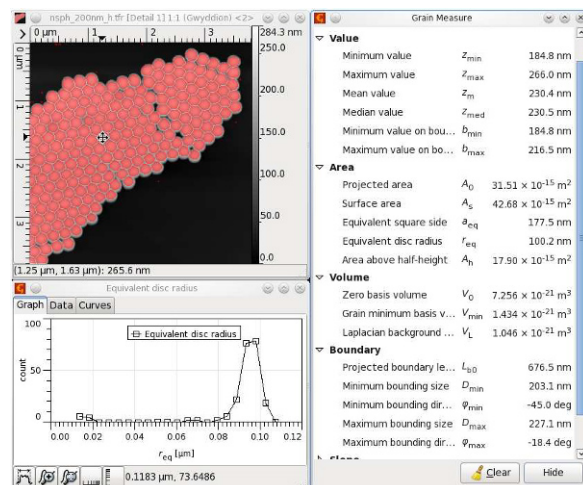


Figure 3. Gwyddion screenshot showing AFM measurements of a layer of agglomerated polystyrene nanoparticles on a silicon surface, masked by the watershed algorithm and evaluated using a particle-equivalent projection method (graph). An individual particle's properties for the particle marked by the mouse pointer are displayed as well.

2.5. Metrology support

Gwyddion also features basic support for metrology in SPM. The key objective is to provide not only the values of direct or statistical results, but also their uncertainties. This process is not straightforward if we use an SPM [6]. First, all the uncertainty sources have to be determined or estimated. Second, uncertainties have to be propagated to the results using uncertainty propagation laws.

The basic idea for uncertainty propagation in Gwyddion is to determine positioning-system errors (positioner-error function) and successively extract the local uncertainties in the full volume that can be reached by the positioning system (including also other uncertainty sources that are not necessarily of a local nature) [16]. In this way, we obtain volume-calibration data. This is in contrast to the more traditional method of measuring separate errors for each axis complemented by further cross-talk measurements and combining them to obtain an overall uncertainty leading to a possible overestimation of the uncertainties.

After obtaining volume-calibration data, each pixel of the SPM image or profile is connected to the known measurement uncertainty in all directions. The volume-calibration data are then used for uncertainty propagation in the data-evaluation algorithms, and results are amended by their uncertainty. Moreover, the volume-uncertainty data processing system in Gwyddion can be used also to compare different microscopes or their parts, like the positioning stages.

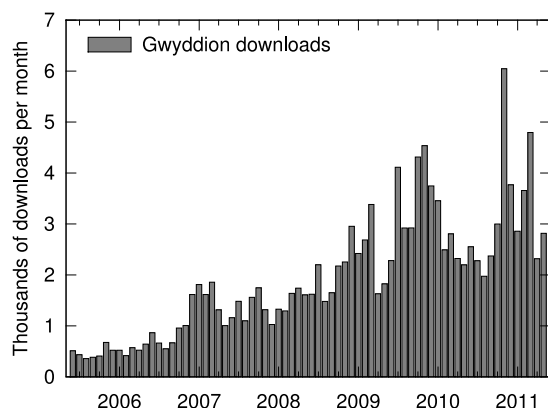


Figure 4. Gwyddion downloads from SourceForge per month (data source: sourceforge.net).

3. Results and future plans

Even though we do not collect statistical data concerning Gwyddion use and spread, some basic information about its use can be deduced from the monthly number of downloads over the years of Gwyddion's development, as shown in Fig. 4. Although an individual download does not necessarily imply an installation or the use of Gwyddion, it can be seen from the graph that there is a clear need for such a software in the community of SPM users.

There are many open questions and tasks for future Gwyddion versions. Besides architectural improvements, there are certain key data-processing tasks that we would like to address in the near future:

- Support for non-equidistantly sampled data in SPM. Increased sampling density in the vicinity of specific surface features (e.g., sharp edges) allows the user to improve surface characterization while keeping the measurement time and amount of data reasonable. Instruments measuring at arbitrary points instead of a regular grid have already been built [9]. However, the data obtained from such instruments cannot be processed by the current generation of SPM software.
- Uncertainty propagation into all results using Monte Carlo propagation mechanisms.
- Extending the set of operations that support masking. Especially the calculation of one-dimensional statistical characteristics, such as the autocorrelation function and the power-spectrum density function of arbitrarily shaped areas would be useful.

- Development of faster statistical-analysis methods based on frequency analysis.

4. Conclusion

Over the course of several years of development, a fully functional software for SPM data analysis has been created featuring a unique modular structure and cross-platform availability. Due to its architecture, it can be used as a standalone software, or its libraries can be employed for other custom software in the contexts of data processing, visualization, or acquisition. Gwyddion reaches a large part of the worldwide community of SPM users.

We have experienced that the open-source structure and the detailed documentation of all the functions lead to fast bug detection and many suggestions for improvements from users. Even though feed-back usually focuses on a specific feature, and even though surface physicists are often inexperienced programmers, many improvements have been brought to our attention by users. This user feed-back ranges from minor bug reports, feature requests, and documentation improvements to the testing of file-loading modules and complex data-processing functions.

This can also be understood as one part of the quality-assessment process, which is in this form lacking for proprietary software. The open-source approach, together with tight coding and testing rules can therefore partly substitute costs that would be necessary for monitoring the quality of such complex software.

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